

Defining ECLSS Robustness for Deep Space Exploration

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Human exploration of deep space will require Environmental Control and Life Support Systems of increasing dependability as mission duration and distance from Earth increases. As crews travel to distant unexplored environments, designers will need heightened confidence in life support availability under increasing levels of uncertainty and risk. Variation in system performance, environmental conditions, resource consumption, waste generation, and even mission characteristics will lead to unexpected responses, increased likelihood of failures, and even design obsolescence. The cost of system failures will also rise, due to launch mass and volume constraints, time and cost of resupply, and reduced ability to abort to Earth. If not accounted for early in design, this increased risk and cost of uncertainty might preclude the economic feasibility of human deep space exploration. The choice of an ECLSS architecture will include many different combinations of technologies that fulfill the functions of atmosphere revitalization, waste removal, and the provision of food and water. The difficult question facing ECLSS designers is how to optimize for the system performance and cost effectiveness necessary for mission feasibility. How do we reduce mass, energy usage, and waste while also ensuring availability of critical life support functions? An optimal architecture must take into account many facets of system performance, such as quality, safety, reliability, and maintainability. Exploration life support systems must continue functioning under harsh conditions, for long durations, without support, and with limited resources or resupply. This review discusses sources of uncertainty in deep space ECLSS design and potential impacts on system functionality. The concept of *robustness* is proposed to characterize and improve ECLSS performance in off-nominal conditions and abnormal operation. The characterization of ECLSS robustness will lay the foundation for future work in identifying design features and practices for sustaining operation in the face of the uncertainty of deep space missions.

Nomenclature

C	=	index of system closure
DDT&E	=	Design, Development, Test, and Evaluation
ECLSS	=	Environmental Control and Life Support System
EMAT	=	Exploration Maintainability Analysis Tool
ESM	=	Equivalent System Mass
EVA	=	Extravehicular Activity
f(t)	=	failure probability density function
F(t)	=	probability unit fails by time *t*
FMEA	=	Failure Modes and Effects Analysis
FTA	=	Fault Tree Analysis
h(t)	=	instantaneous rate of failure for survivors at time t
ISS	=	International Space Station
LCC	=	Life Cycle Cost
LEO	=	Low Earth Orbit
m	=	mass of stores consumed
M	=	total mass consumed

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MARCEP = Maintainability and Reliability Cost Effectiveness Program
MOE = Measures of Effectiveness
MOO = Multiple Objective Optimization
MRBD = Modified Reliability Block Diagrams
MTBF = Mean Time before Failure
MTTF = Mean Time to Failure
P_{LSS} = probability of catastrophic ECLSS failure
P_{QL} = probability of human fatal mistakes due to life support quality
P_{GMS} = probability of failure of supporting equipment
PRA = Probabilistic Risk Assessment
PrLOC = Probability of Loss of Crew
R(t) = reliability function or survival function
RELISSA = Reliability Environment for Life-Support System Simulation and Analysis
SEBoK = Systems Engineering Body of Knowledge
t or T = time

I. Introduction

THE one common objective of any environmental control and life support system (ECLSS) for human spaceflight is to maintain an environment "suitable for the well-being of men and systems during the mission" in an isolated volume.[1] The ECLSS purpose is to maintain the habitability of the spacecraft. It must provide for the crew's needs, in order to keep them alive (at a minimum) and preferably enable them to be healthy, happy, and productive (to achieve mission objectives).[2] Beyond this commonality however, there is no one-size fits all ECLSS for human space exploration. The specific functional, performance, and operational requirements for ECLSS design are derived from mission characteristics (duration, the number of crew, destination environment, and mission objectives). The primary drivers of ECLSS design are the crew's metabolic inputs and outputs (to which the ECLSS provides a counterbalance), environmental conditions (largely determined by mission destination), and mission objectives (dictating the duration, return time to Earth, ability to resupply, and crew activities). The crew consumes resources (food, oxygen, and water) and produces waste (CO_2, humidity, heat, urine, feces, trace contaminants, trash, etc.) In addition to metabolic needs, the crew requires protection from the hazardous space environment characterized by extreme temperatures, high vacuum, micrometeoroids, reduced gravity, and high levels of cosmic and solar radiation. Other mission characteristics that might constrain the ECLSS design include allowable mass and volume, the expected frequency of extravehicular activity (EVA), expected workload of the crew, surface operations, and distance from Earth for return or abort.

The ECLSS must first satisfy the physiological needs of the crew by regulating the atmosphere, providing potable water, and removing hazards posed by wastes. The next level of priority is to provide for crew safety, comfort, and well-being with food, safety infrastructure, mental and physical health countermeasures, and other human factors accommodations. The primary ECLSS elements are atmosphere management, water management, waste processing, food supply, and crew accommodations. Crew accommodations have not typically been included as an ECLSS element but we propose its future integration into ECLSS design and evaluation. Psychological and physiological countermeasures affect crew wellbeing, reduce the risk of illness and injury, and ultimately improve spacecraft habitability, even if those effects may be more difficult to quantify. Several NASA references define specific performance requirements, baseline values, and constraints for these elements, including the Human Integration Design Handbook,[3] the NASA Spaceflight Human System Standard,[4] and the Life Support Baseline Values and Assumptions Document.[5]

The variety of technology options available to provide ECLSS functions have evolved significantly since the beginning of human space exploration, especially in the areas of atmosphere revitalization and water recovery. The first life support systems flown in space were open loop, providing the necessities to ensure human survival, such as pressurized cabins, chemical (KO_2), or high-pressure oxygen storage, CO_2 scrubbing (LiOH), heat exchangers for temperature control, and wick type water separators to remove humidity. Water was stored on-board, and body wastes were collected and either stored or vented to space. Technology has evolved over the last 50 years towards regenerable or recycling systems that sustain a habitable environment for longer durations, with less resupply mass.

II. ECLSS Architecture Development and Optimization

A system architecture is its organizational structure, including its elements, their relationship with the external environment and with each other, and the system implementation.[6] An ECLSS architecture might include many

different combinations of technologies that fulfill the functions of atmosphere revitalization, metabolic waste removal, and the provision of food and water. The difficult question facing exploration ECLSS architects is how to optimize for the performance and cost effectiveness necessary for mission success. How do we reduce mass, energy usage, and waste while also ensuring consistent availability of critical life support functions? The best architecture is not necessarily a collection of the most individually efficient components, but a collection of components that work together in an optimal way.[7] With such a variety of technologies available, there has been a great deal of effort put into the analysis of various architectures in order to inform decisions for mission planning and establish baselines for more detailed systems design and development.

A. Design Optimization Methods: A Brief Overview

Optimization processes attempt to find the best possible design solution amongst all available solutions.[8] *Design optimization* chooses a set of design features (or a system architecture) from a range of feasible options (called the 'design space') that minimizes or maximizes some objective function, subject to a set of defined constraints. Typically, an optimal architecture must take into account many facets of system performance that are important to its users, such as quality, safety, reliability, and maintainability. Multiple objective optimization (MOO) (a.k.a. vector or Pareto optimization) techniques are used for design problems with more than one objective function. For example, one might want to minimize cost, while also maximizing product reliability. If objectives conflict with one another (as they often do), there can be no design solution that optimizes all of them at the same time. The "Pareto optimum solution set" includes the combination of architectures for which any further improvement in one objective will result in degradation of another objective. The Pareto set lies along the boundary of the feasible design space, known as the Pareto front, illustrated in Figure 1. The "optimal" design then becomes a choice along the Pareto front, based on the relative importance of the objectives. Design optimization requires definition of many parameters and processes: a feasible design space or set of design parameters; objective functions (i.e. evaluation criteria) for comparing design solutions; methods for assessing the objectives for a specific design; methods for searching through the design space; and methods for choosing an optimal solution or set of solutions.

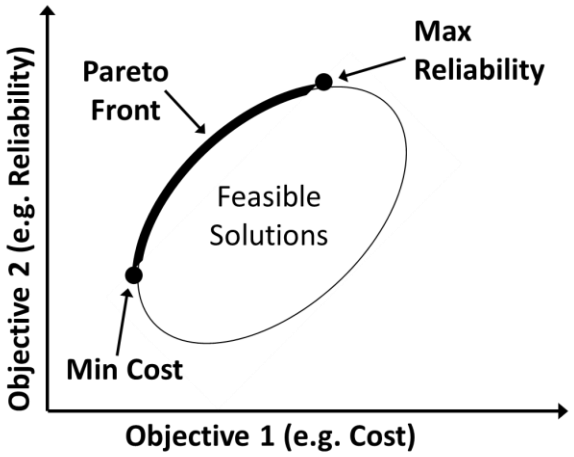


Figure 1. Pareto front example.

Historically, spacecraft design has been heavily dependent upon the use of heritage systems. Existing flight-proven designs are modified to fit new requirements, achieving feasibility but not optimality.[9] The number of potential design variables and objective functions for a spacecraft configuration (and all of its subsystems) are prohibitive for optimizing across the entire design space. In addition, the response surfaces for spacecraft performance are often very complex and require significant user involvement to develop.[10] For these reasons, spacecraft architects have sought to reduce the size of the design space and therefore computation resources using metaheuristic methods, such as genetic algorithms or simulated annealing.[9,10,11,12] ECLSS is a complex spacecraft system, with highly coupled sub-systems and components, which might benefit from the application of multi-disciplinary and metaheuristic methods during architecture development. Several studies have demonstrated the use of optimization algorithms for ECLSS technology trades, configuration comparisons, and system sizing.[13,14,15,16,17,18,19]

B. Feasible ECLSS Architecture Identification

A feasible design solution is one that satisfies hard constraints that are essential, as opposed to soft constraints that users or designers would like to satisfy but are not essential.[8] Early identification of such hard constraints limits the design space for subsequent optimization. Technical performance, as well as cost and schedule constraints typically drive ECLSS feasibility studies during early mission planning.[20] At this stage in design, steady state, back of the envelope estimates of system performance and cost are sufficient to identify candidate architectures and reduce the design space before more detailed iteration.[21] Technologies chosen for exploration missions might borrow directly or evolve from those used aboard the International Space Station (ISS), or develop from recent technical advances to address changing needs and demands.[22] Even in the face of cost, schedule, and mass constraints, the use of heritage systems may not meet the technical requirements for longer duration missions. To aid in feasible architecture

identification, there is a wealth of analyses over several decades that have resulted in rules of thumb for life support system design.[21]

After identifying a top-level ECLSS architecture, designers proceed to more refined architecture definition, choosing the size and number of individual components, defining their interfaces, their configuration, and identifying lower level sub-system requirements and elements. Just as before, the first step is to identify combinations of components that are feasible, satisfying hard constraints. More detailed architecture development depends upon the specific performance or operational requirements derived from mission objectives and constraints (such as available volume, geometric limitations like spacecraft diameter, safety thresholds, structural load limits, temperature limits, etc.), as well as soft constraints such as reliability, maintainability, or usability.

C. Performance Metrics for ECLSS Comparison

After identifying the feasible design space designers must then choose the "best" solution. When more than one design option meeting all requirements is available, soft constraints, also known as objectives, guide the architectural comparisons. Soft constraints contribute to a higher quality system and typically become objective functions for design optimization (such as maximizing reliability or minimizing cost). These objectives, often referred to as '-ilities', reflect the ability of the system to meet the users' needs and expectations. Measures of effectiveness (MOE) are the specific metrics used to measure how well the system achieves its objectives (a.k.a. Measures of Performance, Performance Metrics, Performance Indicators, or Figures of Merit). Typical objectives address the systems effectiveness and its cost, to varying degrees. However, cost and effectiveness criteria usually conflict requiring multi-objective optimization techniques. Ideally, all vested stakeholders (designers, users, investors, etc.) are involved in choosing and prioritizing these criteria, herein referred to as *performance metrics*. Designers can define an apriori utility function based on the relative importance of each metric or present decision makers with a set of Pareto optimal solutions. For ECLSS evaluation, analysts must first define successful operation as well as cost. ECLSS architecture trades typically compare mass, reliability, and closure. For long missions stored consumables, levels of closure, and spare parts to improve reliability are major drivers of system mass.[20,23] The most common ECLSS performance metrics are described in detail below.

1. Equivalent System Mass (ESM)

ESM is NASA's standard measure for evaluating ECLSS system cost, and it is by far the mostly widely used metric for ECLSS trade studies in technical literature. Originally developed by Ref. 24, ESM includes the mass of a system and associated infrastructure given a specific mission location, duration, and crew size.[25] ESM approximates system launch cost as a sum of actual mass (initial infrastructure plus consumables) and equivalent masses of volume, power, cooling, and crew time. Ref.'s 25-26 define the metric in detail and provide guidance for implementation. If the technologies under comparison have the same function at the same quality, reliability, and safety, the best option becomes that with the lowest ESM.[25] Researchers must apply their own expertise to adjust calculations for equal quality, reliability, and safety by adding mass for spare parts, consumables, tools, design modifications, etc. ESM has the advantage of allowing comparison of multiple systems with one utility function that incorporates all primary factors affecting system mass. If used correctly, by incorporated mass to achieve equal reliability, it also accounts for system performance. Other advantages of ESM are its applicability to systems, sub-systems, and components; understandability and ease of computation; and objectivity and precision allowing meaningful technology comparison. ESM has some disadvantages (listed below), however, that make it a better criterion for early architecture (Phase A) development than for detailed (Phase B/C) system design optimization. ESM provides a rough order of magnitude, best used in early mission planning.

1) *Large Design Space:*[7] Because the ESM of an integrated system is not necessarily equal to the summed ESM of its parts (in the case of multifunctional components), all possible combinations of technologies must be traded against one another, resulting in a large design space for optimization.
2) *System Scaling and Stability:*[27] Analysts must take care in system scaling, by sizing equipment to achieve steady state flow rates through the system and mass balance. The validity of ESM is therefore dependent upon the quality of the sizing analysis.
3) *Crew Time Bias:* Though the addition of crew time mass is an excellent way to incorporate the maintenance burden of a system into one utility function, the calculation of the crew time cost factor may create an unfair bias. A 50% reduction in available crew time might not necessarily require twice the system mass.
4) *Location Dependence:* Propulsion mass and hence launch cost changes with mission location. Analysts must only compare systems destined for the same location with the same propulsion type.[21]
5) *Other Life Cycle Costs:* Analysts should not use ESM as a standalone measure of system cost. ESM is proportional to launch cost, but it does not include costs for design, development, testing, and evaluation (DDT&E) or operations.[21]

2. Life Cycle Cost (LCC)

Because costs for development, launch, and operations vary depending on the mission destination and duration, LCC is a more inclusive and accurate cost estimate than equivalent mass.[21] According to the Systems Engineering Body of Knowledge (SEBoK), life cycle cost is "the total cost of implementation and ownership of a system over its useful life. It includes the cost of development, acquisition, operation, maintenance, support, and, where applicable, disposal."[6] LCC is a standard systems analysis metric used by NASA in mission design and throughout the aerospace industry.[21] For space exploration missions, LCC will include DDT&E, launch and emplacement, and operations. Ideally, system disposal or retirement should also be included in LCC. Ref. 21and Ref. 28 provide an overview of methods for estimating the different components of LCC.

3. Closure

Closure reflects the ability of an ECLSS to continue functioning without resupply, defining its autonomy.[14] The Index of System Closure (*C*) characterizes the portion of expended substance that is recycled (not stored) per unit time: $C = 1 - m/M$, where m is the mass of stores consumed, and M is the total mass consumed.[29] Designers might treat closure as a goal (rather than a requirement), as complete closure (C equal to one) is not practical to achieve.[14]

4. Stability

For *stability*, a steady state in mass balance must exist for each element in the system, in terms of O_2, CO_2, H_2O, etc. Non-zero mass into the ECLSS must equal mass out or else deadlocks will result, eventually leading to system failure. Ref. 14 cautions that the index of closure can only be meaningful if there exists a settled stationary state and the total mass of the system is constant. Systems of differential equations describing mass flow dynamics between elements allow stability analysis. ECLSS analysts often consider stability in the context of biological life support systems, but it is also applicable and just as important for physico-chemical (P/C) systems.

5. Reliability

Reliable functioning ensures the safety of human life under space mission conditions.[30] Mass based optimization assumes that a higher degree of recycling (or closure) is desirable for long duration missions. However, due to the inability to get replacement parts or other resupply from Earth if failure occurs, reliability may become more important than mass. Recycling (as opposed to storage) may not be the most reliable means of providing for the crew's life support.[31] The need for extreme ECLSS reliability for long duration missions is hardly a new revelation. In 1968, Ref. 32 stated,

> *"Reliability and maintainability problems confronting environmental control/life support systems for long duration space flight present the designer with one of the most challenging tasks in the wide array of still unproved approaches to the design of manned space systems."*

The Systems Engineering Body of Knowledge (SEBoK) defines reliability as "the probability of a system or system element performing its intended function under stated conditions without failure for a given period of time."[6] A *failure* is the inability to perform as intended. There are several metrics for reliability, including mean time to failure (MTTF) for non-repairable systems, mean time between failures (MTBF) for repairable systems, failure rate, or probability of failure. A reliability function $R(t)$, also known as the survival function, is the probability that a unit (or system) survives beyond time t: $R(t) = 1 - F(t)$, where $F(t)$ is the probability that a unit will fail by time t.[33] $F(t)$ is the cumulative distribution function of failure probability density function $f(t)$, defined for t=0 to infinity. Failure rate, $h(t)$, is the instantaneous rate of failure for survivors at time t during the next instance of time: $h(t) = f(t)/R(t) = f(t)/(1-F(t))$. The integral of $h(t)$ from 0 to time t is known as the cumulative hazard function.[33]

Ref. 31 defines ECLSS reliability as the probability of loss of crew (PrLOC) which is equal to the failure rate per day for life-threatening failures times the mission duration in days. The author suggests that PrLOC due to life support failure must be less than 1/1000 for all mission durations, such that the allowable failure rate per day decreases with mission duration. For a 3-year mission, this would require a daily failure rate on the order of 1E-6 for ECLSS sub-systems or components, termed *ultra-reliablity*.[34]

Reliability Estimation: It is difficult to observe failures in testing, especially for highly reliable systems; hence, reliability engineers have developed modeling techniques to estimate failure rates from test data. Accelerated testing methods test components at high stress levels for failures to occur. Time to failure under normal use can then be predicted as a function of operating stress applied in testing. System reliability is determined as a function of the individual failure rates of the integrated components and their configuration. System models include combinations of series, parallel, standby, or R-out-of-N components. In a series model, one component failure leads to system failure. In a parallel model configuration, all components must fail before the system fails. A standby configuration contains an inactive redundant component that steps in if the active component fails. An R-out-of-N system requires R of the

total N components to be functioning for the system to be operational. Reliability prediction error increases with the number of external factors upon which system performance depends, due to the interaction between factors.[35]

Though publications on life support design have strongly emphasized the need for improved reliability for deep space missions, the number of reliability based trade studies in the literature is small relative to the large number of published ESM analyses. Ref. 32 presented an analytical technique for optimizing the type and number of spare components to achieve a system reliability goal, by iteratively choosing components with the highest Δreliability/Δweight. With this method, a spares weight of 30-40% of single-thread sub-system weights achieved 90% chance of mission success. In a similar study, Ref. 36 developed the Maintainability and Reliability Cost Effectiveness Program (MARCEP) to analyze several manned space missions and optimize mass in terms of the combination of redundancies (parallel and standby) and spares needed to achieve a reliability goal. One simulation showed 15,500 lbs. of redundancies and spares would achieve 0.99 probability of mission success for 730 days, which was about 17% of the total weight of the spacecraft systems. Simulation also showed that the time required for ECLSS maintenance was about 43% of total spacecraft maintenance requirements. Ref. 37 analyzed online (parallel) and offline (standby) redundancy designs using modified reliability block diagrams (MRBD) that account for buffering capacity. However, this technique does not account for repair or recovery time during the failure.[38] These studies also assume random failures. They do not account for inherent design flaws that spares may not mitigate.

Repairability (instead of redundancy) is another means of increasing reliability with less mass penalty and less complexity than that of redundant systems. Ref. 32 suggests that in-flight repair actions (rather than replacement of an entire module) offer improved potential for extended reliability and that the ability of the crew to manage ECLSS is essential to successful system performance.

ECLSS reliability analysis tools include Reliability Environment for Life-Support System Simulation and Analysis (RELISSA) developed by the Institute of Space Systems at the University of Stuttgart[39] and the Exploration Maintainability Analysis Tool (EMAT) developed by NASA.[40]

6. Integral Reliability Criterion

Integral reliability[30] is the summed probability of catastrophic ECLSS failure (P_{LSS}), human fatal mistakes due to life support quality (P_{QL}), and failure of supporting equipment such as power (P_{GMS}), in general form:

$$P_{LSS}(mass) + P_{QL}(T) + P_{GMS}(mass). \quad (1)$$

P_{LSS} typically decreases with increased life support system mass (*mass*). P_{QL} is dependent upon quality of life and time (*T*) of stay in the habitat. P_{GMS}(mass) increases with system mass. Integral reliability is an objective quantitative criterion that requires dynamic modeling and simulation to account for the effects of stochastic external disturbances, emergency scenarios, and perturbations in component performance over time.[7] The mass term includes equipment and spare parts, consumables, or other components to improve reliability. The calculation of P_{QL} is much more difficult to estimate, but assumptions could be made about the time dependent cognitive and physical impairment in a degraded environment and the resulting probability of fatal mistake. The mass dependent probability of failure for general mission support systems (P_{GMS}) depends upon the mission scenario.

III. Cost of Uncertainty in ECLSS Design

As crews travel to distant unexplored environments, mission designers will need heightened confidence in life support functionality under increasing levels of uncertainty and risk. The top technical challenges for human spaceflight beyond low Earth orbit (LEO) are to create safety infrastructure, autonomous operational strategies and systems, highly reliable and maintainable systems, and robust systems to deal with the unexpected.[22,41] This paper discusses the notion of *robust* in the context of uncertainty and proposes robustness as a necessary system characteristic that supersedes and includes the other challenges. In order to develop robust systems, characterization of *the unexpected* is a necessary first step.

A. Sources of Uncertainty Affecting ECLSS Performance

To meet the demands of long duration life support, ECLSS technology will undoubtedly become more complex and have longer operational life spans. As system complexity and life spans increase, the likelihood of encountering variability (internally or externally) also increases.[42] Increased variability means increased uncertainty in component reliability, environmental operating conditions and hazards, crew metabolic loads, internal process parameters governing outputs, component interactions, and even mission characteristics. This uncertainty rises with mission duration and distance from earth.

"Uncertainties are things that are not known, or known only imprecisely."[43]

According to the SEBoK, uncertainty is the result of not having accurate or sufficient knowledge of a situation.[6] Ref. 43 identifies two classes of uncertainty: lack of knowledge or lack of system definition. Many random events are *probable* (frequent and foreseen, statistically characterized events with relatively low impact). Other random events are *possible* (foreseeable events that occur less frequently but have higher impact). While some events are only *plausible* (very infrequent, very high impact rare or "black swan" events that may be unforeseeable by subject matter experts and are difficult to statistically characterize). Space is characterized by many uncertainties, natural and human induced, which might affect system functional performance, making theoretical reliability analysis difficult.[44] Historical data and empirical evidence exists for the characterization of some, but not all sources of uncertainty. That does not mean however, that designs cannot accommodate the unknown. Designers can bound realistic possible ranges of environmental conditions, or types of rare events. Once this uncertainty is characterized, or at least bounded, a designer must then attempt to predict impact on ECLSS performance and on crew-wellbeing. How much performance degradation can be tolerated and for how long? What are tolerable conditions that the crew could be forced to operate in during an emergency? How does ECLSS performance under unexpected conditions affect the choice and optimization of system architecture?

1. *Component Performance Definition*

Component Failure Rates: Since component failure rates are empirical, treating failure rates as uncertain parameters may avoid unnecessarily high redundancy that results in heavier systems.[11] Component failure rates are variable due to differences between individual units that result from imperfect manufacturing, differences in storage conditions and environmental exposures, material imperfections, etc. Failure rates are also time dependent and reliability uncertainty grows over time, due to the expense of collecting failure rate data over long durations.

Component or Sub-system Process Rates: The low technology readiness or flight heritage of newer regenerable technologies results in higher uncertainty in performance parameters such as recovery rates or process efficiencies, especially over time of use. This is especially true for biological components. Uncertainty in forecasting system states, even if the mathematical description of state dynamics is well defined, occurs due to variability in the responses of individual organisms.[14] Variation of process rates between individuals applies to the human crewmembers as well (e.g. metabolic rates, consumable utilization, and waste generation). Higher cycling rates are typical within a space habitat, amplifying and propagating the effects of these uncertainties throughout the system.

Resource Requirements for Mass Estimates: ESM estimates for less mature technology are less accurate when performance and resource requirements are uncertain, and therefore must be adjusted based on technology maturity.[45]

Uncertain Effects of Space Environment: Experience with regenerative life support technology on the ISS has shown reduced gravity or microgravity to have unanticipated adverse effects on component performance. The radiation environment in LEO and beyond is difficult to replicate on Earth as well. Ground testing may not reveal unanticipated effects of space radiation on components. Only long duration flight-testing can verify component performance in the intended operating environment.

2. *System Definition - Plant Model*

Integrated System Performance: Fundamental relationships between system performance, component states, and environmental conditions are ill defined, especially for systems that include biological components or immature technology. System performance predictions require dynamic models of non-linear component interactions that drive emergent system level behavior. Benchtop component level experiments provide empirical model coefficients that are less certain than those derived from full-scale experiments. However, experimental data for integrated systems is expensive to collect.

Dynamic Process Rates: Some mass and energy exchange processes have stochastic behavior, such as human metabolism, plant growth, heat exchange, gas diffusion, etc. Therefore, mass and energy flow rates throughout the system will exhibit transient behavior that may or may not be consistent or predictable. Transients in mass and energy exchange rates may result due to ill-defined complex component interactions.

3. *Metric Uncertainty*

Metrics are approximations or indicators of an objective. They also contain inaccuracies in their formulation and in the level of diligence applied to their calculation. For example, ESM inaccuracies result from location and vehicle dependent power, volume, or heat rejection conversion coefficients.

4. *Operating Environment and Usage*

Internal Operating Environment Instability: ECLSS performance controls the internal spacecraft environment, which in turn affects ECLSS performance. Uncertainty in system, sub-system, and component performance for ECLSS presents uncertainty in the internal operating environment of the spacecraft. Transients in gas composition, pressure, temperature, humidity, contamination, etc. may occur that could result in unexpected system loads.

Actual In-flight Utilization and Procedures: Ref. 32 observed that differences in in-flight utilization and maintenance procedures result in large variability for MTBF in airline systems (more so than variability in quality between manufacturers). System usage in flight by the crew is a source of uncertainty that influences reliability of system performance. In this way, the crew is a system element upon which ECLSS performance depends.

Boundary Conditions: Variable mass and energy exchange rates between the spacecraft and the environment include atmospheric leakage, energy inputs, and energy loss.

External Operating Environment Uncertainty: Estimates of the expected radiation environment, micrometeorite exposure, temperature, and even gravity come from theoretical and empirical models based upon remote observations. Predictions of the anticipated space environment are likely quite good for the purposes of system design. However, as exploration missions take spacecraft into less traversed environments, the likelihood of unexpected environmental variability increases. Because the space environment is so extreme, the consequences of this uncertainty may be equally extreme, even if the uncertainty is small. The purpose of exploration is to gain knowledge. Thus, uncertainty in what we will find during the journey and when we get there is inherent to that exploration.

Rare Events: Rare events are those with potentially widespread, severe impact that occur with low frequency, such as earthquakes, asteroid impacts, solar flares, etc. In aerospace engineering, rare events also apply to less catastrophic events that may happen at low probabilities of less than 1E-6.[46] The likelihood of rare events (e.g. a large dust storm on Mars or a large meteor impact on an asteroid) increases as we move further into unexplored, little studied environments, simply because probability estimates have limited accuracy.

5. Mission Characteristics

The mission itself may be a source of uncertainty for exploration life support system design. More far reaching mission objectives may have inherent uncertainty in ECLSS design drivers such as mission duration, resupply intervals, and crew size. Simply the passage of time during long duration exploration increases the likelihood of mission changes in response to unanticipated events, such as funding shortages, launch vehicle problems, changes in science objectives based on new knowledge, etc. Even during early mission planning, system designs must be flexible to mission uncertainty. As the specific goal of exploration missions continues to shift (e.g. the moon, asteroids, or Mars), it becomes necessary to maintain a *flexible path* philosophy by designing architectures that are capability driven and encompass all mission possibilities.[47]

B. Cost of Uncertainty

Failures generate costs (during development, production, and operation), and failure costs increase the later they occur in the product life cycle. The cost of failure can be direct, such as the cost of production rework and scrap, and indirect, such as the cost of schedule delay or loss of sales.[48] Failures can even lead to product recalls and litigation in the case of injury or death.[48] Strategies to reduce failures and hence their cost have a price, called the *cost of quality*. Quality engineers attempt to find the minimum cost of quality, which may or may not occur before reaching 100% reliability. Modern experts tend to think that by eliminating the root causes of failure, the savings will always be greater than the cost of prevention.[48]

For space life support, increased uncertainty may result in unanticipated system behavior, increasing the likelihood of component faults, system failure, or even design obsolescence. The impact of safety critical system failures and the cost of their prevention also rise with mission distance and duration, due to launch mass and volume constraints, time and cost of resupply, and reduced ability to abort to Earth. Without multiple vehicles or abort capability, a system failure could result in a loss of crew and an entire campaign, rather than just a loss of vehicle or mission. Ref. 49 cautions that future life support system complexity (large mix of technologies with large variety and numbers of components) will lead to lower reliability and higher maintenance requirements than other systems.[36,49] A number of studies for crewed space missions indicated that ECLSS generally requires more scheduled and unscheduled maintenance than any other system, and that this has been the case since the first crewed space missions.[36,50]

Traditional methods for improving reliability (such as incorporating redundancy or carrying spares) to achieve acceptable risk may also be too costly or ineffective to ensure system operability over long durations and uncertain conditions.[51] The reduced ability to resupply or abort heightens the need for systems that can be repaired and maintained on-board. Spares will have to be flown with the spacecraft or pre-positioned, requiring a large mass and volume for storage and transport. In addition, a greater number of spares is required to accommodate increased uncertainty in component reliability.[51] Other approaches to reduce the cost of uncertainty will be required, such as component commonality, functional redundancy, reduced complexity, and in-situ manufacturing.[51,52,53]

IV. ECLSS Robustness: Performance under Perturbation

A. Robustness, Reliability, Resilience, and Survivability: What is the Difference?

1. The Change Related '-ilities': Establishing a Semantic Frame of Reference

There is a plethora of design objectives (or '-ilities') for engineered systems related to performance under uncertain, adverse, or unanticipated conditions. This list includes (but is not limited to) robustness, reliability, resilience, survivability, vulnerability, availability, supportability, maintainability, susceptibility, adaptability, recoverability, stability, and changeability. Depending on the industry and application, the terms may have different or often overlapping meanings, with different relationships to one another. A comprehensive, yet concise overview of these "change-related '-ilities'" is given in Ref. 42.

In the context of ECLSS design, any effort to improve confidence in life support effectiveness (at maintaining a suitable, habitable spacecraft environment) under increasing levels of uncertainty and risk will require a common semantic frame of reference. This allows designers and stakeholders to use a common contextual language intelligibly,[54] in order to adequately express and understand ideas and concepts. Characteristics that may (or may not) contribute to the system effectiveness in an uncertain changing context must be clearly defined, before they can be incorporated into the design process. A review of literature on change-related '-ilities' reveals four emergent terms that best describe system effectiveness in a dynamic uncertain environment, or *performance under perturbation*: *robustness, reliability, resilience, and survivability*. With the exception of reliability, the definitions of the other three vary greatly in literature, even within specific industries. Some say that resilience encompasses robustness, and others say the reverse, that resilience is a part of being robust. Many use the terms interchangeably, while others maintain that they are two distinct and even mutually exclusive system characteristics. The same is true for survivability and resilience. In this work, we propose that robustness is actually an over-arching characteristic that captures reliability, resilience, and survivability.

2. Definition of Robustness

The Oxford dictionary defines a robust system as "able to withstand or overcome adverse conditions." The Merriam Webster dictionary defines robust systems as "capable of performing without failure under a wide range of conditions (in software)." Random House defines it as "strong and effective in all or most situations and conditions." In technical literature, much like resilience, definitions of robustness for engineering systems vary. Some definitions treat robustness as distinct from resilience, emphasizing an unwavering resistance to disturbance or perturbation, without any change in the system.[55,56] The Taguchi method, known in industrial quality engineering, defines a robust design as one that delivers a strong signal regardless of external or internal noise.[57] "Robust products can perform intended functions under realistic downstream production and disturbances due to usage, without controlling the sources of disturbance."[57] Many other definitions extend robustness to include the ability to maintain output, function, or features regardless of changes in the system or changes in context.[58] Others extend it to allow both passive and active means of maintaining function:

> *"Robustness is capacity to cope with ordinary and occasional disturbances by avoiding exposure, being resistant, or responding/recovering after disruption."*[59]

Proposition 1: The concept of *robustness* is proposed to characterize and improve ECLSS performance. We suggest that ECLSS robustness encompasses all aspects of its mission effectiveness. ECLSS function, or purpose of operation, is to maintain habitability of the spacecraft environment. Therefore, ECLSS *robustness* is its *ability to maintain habitable conditions for crew survival and productivity over the mission lifetime 'under a wide range of conditions'* (Merriam Webster definition). This wide range of conditions includes ordinary usage, finite occasional disturbances or disruptions, and longer term, sustained changes in the system or mission context. Robustness includes the ability to function during expected circumstances, as well as in a changing environment, or during unanticipated events. It allows for limited degradation of performance as long as crew survival *and* their ability to perform mission work are not compromised. It includes contingency operations for crew survival during adverse events and emergencies.

Proposition 2: Three separate system characteristics that contribute to ECLSS robustness are reliability, resilience, and survivability. ECLSS robustness is the insensitivity of performance (i.e. maintaining habitability) to

1) Random expected failures and conditions (reliability),
2) Foreseen but unexpected deviations in conditions or disturbances (resilience), and
3) Unforeseen disturbances or adverse events (survivability).

Each of these characteristics has varying applicability, effectiveness, and cost depending on the mission circumstances. By definition, robustness widens the conditions in which the system performs acceptably beyond

expected (i.e. nominal) to unexpected (less likely) perturbations and disturbances. By its stochastic nature, reliability itself (or likelihood of random failure over the life cycle) is an uncertainty. Therefore, robust encompasses reliable. At a higher level of robustness, the system is resilient to unexpected but foreseeable deviations in conditions. At the highest level of robustness, the system also demonstrates survivability to unforeseeable but plausible conditions and disturbances. Resilience includes the ability to absorb, withstand, or recover from events outside of those it is designed for; and possibly even adapt or change itself to avoid future adverse events. Survivability is the ability to deliver essential services in the face some adverse, typically catastrophic event. Figure 2 depicts reliable, resilient, and survivable as three distinct layers of robustness across a continuum of uncertainty. The further characterization of ECLSS robustness and contributing characteristics will lay the foundation for future work in identifying design features and practices for sustaining operation in the face of the uncertainty of deep space missions.



Figure 2 Robust Characteristics

3. Definition of Reliability

Reliability is herein defined according the SEBoK definition: "The probability of a system or system element performing its intended function under stated conditions without failure for a given period of time."[6] The distinguishing aspect of the SEBoK definition of reliability is performance under "stated conditions" for a given period. Stated conditions are sometimes called nominal. Nominal operating conditions may be static, or they may include some known, well-characterized variability that a component is designed to tolerate, often reflected in specific design requirements. Reliability includes the probability of random failures but also the system availability (with respect to repair and maintenance). In computer science and software engineering, the term dependability (also known as RAMS) is sometimes used. RAMS includes reliability, availability, maintainability, and security. However, in traditional quality engineering, reliability is sufficient to encompass these additional aspects. Hence, for this research the term reliable is chosen, as defined above, and may be treated synonymously with dependable.

4. Definition of Resilience

The Oxford dictionary defines resilience as "the capacity to recover quickly from difficulties; toughness" and "the ability of a substance or object to spring back into shape; elasticity." The definition of resilience varies widely in the engineering literature. The Department of Homeland Security[60] defines resilience as the ability to adapt to changing conditions and prepare for, withstand, and rapidly recover from disruption. Ref. 61 defines resiliency as a superset combining survivability and recoverability. Distinct attributes of resilience include flexibility or elasticity to withstand unexpected failures; the ability and time to recover function after failure or adverse events; and the ability to change or adapt to deal with stresses or changes. Another distinguishing feature is that a service or capability disruption has occurred. Resilience typically involves active responses (flexibility, recovery, and adaptation), as opposed to passive resistance to adversity or perturbation.

5. Definition of Survivability

The concept of survivability has evolved from the domain of military systems and has become an important metric in design of public networks, infrastructure systems, and space systems.[61]

> *Survivability may be defined generally as the ability of a system to minimize the impact of a finite disturbance on value delivery, achieved through either the satisfaction of a minimally acceptable level of value delivery during and after a finite disturbance or the reduction of the likelihood or magnitude of a disturbance.*[62]

Distinctive features of this definition are the magnitude of impact of *finite* disturbances on the system, in terms of either service disruption or impairment. The Department of Defense defines vulnerability and susceptibility as subsets of survivability.[63] Resistance and stability are two potentially contributing factors to survivability. In many texts, resistance defines robustness, in the context of output insensitivity to variations of inputs or other environmental

factors. *Resistance* is the degree to which a variable changes, following a perturbation.[64] Similarly, stability is sometimes synonymous with robustness and resistance to change. Stability is the ability to return to an equilibrium state after a temporary disturbance.[65] Stability contributes to resistance, and resistance may contribute to both resilience and survivability.

B. Need for an ECLSS Robustness Metric

Robustness becomes increasingly important as missions venture further from Earth.[27,52] Exploration life support systems must continue functioning under harsh conditions, for long durations, without support, and with limited resources or resupply. Due to the increased uncertainty and risk of deep space exploration missions, the ability to maintain function in abnormal, off-nominal conditions as well as nominal conditions is increasingly important. However, ECLSS performance to date falls short of the extreme reliability requirements evidenced by actual failure rates as well as on-board maintenance requirements.[32, 47, 50, 66]

Ref. 34 suggests that long duration missions must not only have ultra-reliability on the order of 1E-6, but that reliability must also have *low variation*. Traditional reliability analysis does not account for many factors that affect system performance, including component interactions, tolerances, possible design errors, failure rate uncertainty, operator error, unexpected environmental conditions, partially failed states, buffering capacity, or maintenance or repair strategies.[34, 37, 51]

Ref. 7 points out that it is important to find a "maximally stable system" that can cope with emergencies without losing stability. The authors describe stability as the ability of the system "not to exceed certain thresholds by itself and the ability to return to a nominal functioning (all values within defined thresholds) after a system disturbance." The objective for ECLSS performance should expand beyond reliability, or even ultra-reliability, to robustness, in order to reduce the cost and risk of uncertainty for deep space exploration. Many studies have proposed methods for ECLSS reliability estimation and expanded techniques to consider redundancy, maintenance, repair, and buffering. However, the methods reviewed thus far do not account for the range of possible events and disturbances that could affect performance.

If robustness is a new optimization objective, then it must be quantifiable for design evaluation and comparison. A quantifiable robustness metric serves three purposes in ECLSS design: design evaluation, design comparison for trades and optimization, and design improvement by allowing identification of robust design features.

> *"Measurement is the first step that leads to control and eventually to improvement. If you can't measure something, you can't understand it. If you can't understand it, you can't control it. If you can't control it, you can't improve it." - H. James Harrington*[67]

ESM is a valuable measure of system cost, but should not be the exclusive criteria for system evaluation, especially for missions with increasing demands for robustness. In theory, analysts use ESM to compare systems of equal reliability, which is extensible to equal robustness. However, it is unlikely that ECLSS designs have equal reliability or robustness in practice, due to diligence applied in analysis and the complexity of system designs. If robustness is the more important objective, then it should be in the numerator, rather than the denominator (i.e. performance per cost instead of cost per performance). This brings more focus in design on ways to improve performance, and less focus on mass. Would mission planners want the lowest ESM system that meets a desired, acceptable reliability or the most robust system for an acceptable ESM? Ref. 32 points out that systems procured with a "best performance for a reasonable cost" philosophy, rather than a "lowest cost for a specified performance" approach tend to achieve higher reliability. ESM measures focus system development towards the latter approach, instead of establishing an acceptable cost threshold and designing towards the maximum possible performance.

Cost and performance are conflicting optimization objectives. Ideally, analysts should evaluate them separately and present a Pareto optimal set of solutions to decision makers, allowing for shifting cost vs. performance priorities.

C. Suggested Characteristics for a Robustness Metric

Habitability vs. Crew Survival: Current performance metrics focus on failure and crew survival but do not quantify habitability in terms of productive crew time available, which is ultimately the purpose of the life support system. If the intention were only to keep the crew alive, there would be no requirements for crew comfort. A robustness metric might extend the expectation of performance to include productive capacity of the crew, not just the probability of loss of crew (PrLOC). Ref. 7 suggests that astronaut crew time for science [or other mission objectives] is the most important commodity that ECLSS provides, and that an ECLSS optimization goal is to maximize productive crew time available. Disturbances in ECLSS performance accumulate the loss of crew time, including time spent on

maintenance, time lost due to degraded health [including fatality], time to repair the system after a failure, and time lost due to crew error because of malnutrition or physical stress.[7]

System Levels of Abstraction: Systems have hierarchical levels of abstraction, such as components, assemblies, sub-systems, integrated systems, and systems of systems.[54] It is reasonable to expect that a robustness metric be applicable to all levels of system abstraction. In other words, robustness of a valve should have the same definition as robustness of a heat exchanger, or robustness of an air revitalization system. However, improving performance at a lower level may not improve performance at the top level. A collection of the most robust components may not result in the most robust integrated system. Emergent system level properties occur as the result of architectural features, component interfaces, and relationships such as feedback loops.

Design Fidelity: A robustness metric should be applicable to varying degrees of design fidelity, allowing analytical rough order approximations during preliminary architecture trades, as well as high fidelity estimates for more detailed designs (through numerical simulations and/or testing).

Practicality for Use: A metric must be simple enough that it is practical for use by designers and analysts. It must be understandable, relatively easy to compute, and based on familiar, operationally relevant design features.

Relevance and Objectivity: While a metric should not be so complex that it is impractical for use in design, it should include enough detail to provide meaningful, relevant information. It should be precise and objective enough to allow useful technology comparisons and verification through testing. It should account for all factors contributing to robustness without double counting. If possible, the metric should not include subjective weighting factors.

Compatibility with Existing Assessment Tools: The metric should have a generic formulation that allows its incorporation into current toolsets. Though the authors propose that a new robustness metric is necessary, it should be supplemental to or build upon existing reliability assessment tools (such as EMAT), ECLSS system modeling and simulation tools, and optimization algorithms.

Compatibility with All Technology Types: A robustness metric should be equally applicable to biological, physical, and chemical processes; regenerable and non-regenerable systems; and old and new technologies.

A robustness metric must quantify the system's ability to maintain consistent performance (i.e. conditions necessary for crew productivity) in time, under perturbation of state, and in the event of system disturbance (failure or other shock). Mathematically, robustness might extend the reliability function $R(t)$ from the time domain to include other dimensions of state vector $\boldsymbol{x}$, $R(t,\boldsymbol{x})$.

A survey of ecological, complex systems, and network theory will unearth potential robustness indicators for consideration. Promising criteria include integral reliability,[30] multiplicative probability estimation,[35] ecological stability indexes, and the susceptibility index.[68] Ref. 38 and Ref. 40 also suggest the use of Markov chains with Monte Carlo simulation to predict ECLSS resilience and objectively model uncertainty for systems with little failure data.

V. Conclusion

Future research will explore methodology for the design of robust environmental control and life support systems (ECLSS) for human exploration of deep space. Research objectives are to define terminology, define metrics, develop assessment methods, and provide design guidance for improving ECLSS robustness.

Objective 1. Establish Semantic Frame of Reference: A common contextual language is necessary to adequately express and understand the ideas of systems analysis as it applies to ECLSS design. Establishing common terminology has been the primary goal of this introductory paper. Based on extensive review of literature on the change-related '-ilities', the concept of *robustness* is proposed to characterize and improve ECLSS performance.

ECLSS *robustness* is herein defined as the *ability to maintain habitable conditions for crew survival and productivity over the mission lifetime under a wide range of conditions.* We suggest that ECLSS robustness encompasses all aspects of its effective performance (independent of cost), and that three separate system characteristics contribute to ECLSS robustness: reliability, resilience, and survivability.

Objective 2. Define Measurable Criteria for Quantifying ECLSS Robustness: The most commonly used ECLSS performance metrics (ESM, reliability, closure, stability, and integral reliability) do not sufficiently capture robustness. A metric that brings focus on the ECLSS performance rather than cost will bring increased attention to performance improvement. A robustness metric should address spacecraft habitability, not just crew survival; apply to all levels of system abstraction; apply to all design phases or levels of fidelity; be practical for use, relevant, and objective; and be compatible with existing assessment tools and all technology types. The next phase of research in robust ECLSS design will endeavor to develop a robustness metric incorporating these characteristics. In addition to the methods listed herein, a survey of ecological, complex systems, and network theory will unearth potential robustness indicators for consideration.

Objective 3. Develop and Demonstrate Methods to Assess Robustness: Even with a defined metric, a method of determining its value for a given design will require a consistent and defined assessment process. This will include system definition and modeling, a definition of nominal system performance, definition of the acceptable levels and duration of degraded performance, and characterization of sources and impacts of change or uncertainty that could affect system operation. Methods of bounding, describing, and modeling uncertainty will be necessary for system performance assessment and verification. Existing risk analysis tools such as failure modes and effects analysis (FMEA), fault tree analysis (FTA), and probabilistic risk assessment (PRA) may aid in the identification of likely, possible, and plausible adverse events. Assessment methods will supplement and expand upon existing ECLSS reliability analysis tools such as RELISSA or EMAT.

Objective 4. Provide Guidance for Improving ECLSS Robustness for Exploration Missions: The ultimate goal of this research is to improve ECLSS robustness for long duration distant missions. Metrics and assessment only serve to tell designers how close they are to the achieving the goal but not how to get there. Future work will identify robust design features and recommend best practices for robust ECLSS design. Traditional reliability engineering is not enough to ensure operability under uncertain changing conditions. Efforts to minimize industrial process variation and to ensure the dependability of complex, critical infrastructure (such as communications networks) have resulted in the emergence of survivability analysis, rare events modeling, robust design, and resilience engineering. The application of these specialty engineering disciplines to spacecraft design is somewhat new and their potential for ECLSS design may be yet untapped. A review of systems design literature reveals a variety of characteristics that contribute to system operability during unanticipated conditions, such as redundancy, modularity, failure mode isolation, network connectedness, simplicity, passive control, wide design margins, maintainability, accessibility, interface commonality, capacity for lower levels repair, functional diversity, etc. Lessons learned from the operational history of life support systems to date will also provide invaluable input to improving robustness.

Many studies and reports on ECLSS performance cite the need for robust systems. There has been much progress in the definitions, assessment tools, and design practices for ECLSS reliability. However there has been little attention given to the definition, tools, and practices for robust ECLSS design. Sometimes the word robust is used interchangeably with reliable, resilient, and survivable. We propose that robust encompasses the other three distinct and equally important aspects of system performance. This research will not attempt to replace or improve upon the wealth of system modeling and simulation tools developed by ECLSS analysts. Nor will it attempt to create new optimization algorithms or techniques. Instead, the proposed metrics and assessment methods for robust design will supplement existing modeling and optimization methods, expanding their applicability to a wider range of ECLSS design and operation.

Declaration of Conflicting Interest

In accordance with University of Colorado policies and procedures, and our ethical obligation as researchers, we are reporting that one of the authors, Christine Escobar, has a financial and business interest in a company that may be affected by the research reported in the enclosed paper. She has disclosed those interests fully to the University of Colorado, and she has in place an approved plan for managing any potential conflicts arising from the business interest.

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